

Assignment: Bayesian Probability

Given

Let M = event that Manchester United wins. Let P = event that the pub is packed.

$$P(M) = 0.7$$

$$P(\neg M) = 1 - P(M) = 1 - 0.7 = 0.3$$

$$P(P | M) = 0.9 \quad (\text{probability pub is packed, given United wins})$$

$$P(P | \neg M) = 0.6 \quad (\text{probability pub is packed, given United loses})$$

Required

Find $P(M | P)$ — the probability that United won, given that the pub is observed to be packed.

Bayes' Theorem

$$P(M | P) = [P(P | M) \cdot P(M)] \div P(P)$$

$P(P)$ is unknown, so we find it using the Law of Total Probability. The pub can be packed under two mutually exclusive and exhaustive scenarios — United winning, or United losing:

$$P(P) = P(P | M) \cdot P(M) + P(P | \neg M) \cdot P(\neg M)$$

Calculation

Case 1 — United wins:

$$P(P | M) \cdot P(M) = 0.9 \times 0.7 = 0.63$$

Case 2 — United loses:

$$P(P | \neg M) \cdot P(\neg M) = 0.6 \times 0.3 = 0.18$$

Total probability of a packed pub:

$$P(P) = 0.63 + 0.18 = 0.81$$

Applying Bayes' Theorem:

$$P(M | P) = 0.63 \div 0.81 = 0.7778$$

Interpretation

No — a packed pub cannot be taken as proof that Manchester United won. A packed pub is also fairly likely (probability 0.6) when United loses, since fans may gather to drown their sorrows rather than celebrate. The observation is evidence, not certainty. However, it is more consistent with a win than a loss (relative weight 0.63 vs 0.18), so it raises the belief in a win from the prior probability of 0.70 to a posterior probability of 0.7778.

Answer

Manchester United winning cannot be assumed with certainty. Given that the pub is packed, the probability that Manchester United actually won the game is:

$$P(M | P) = 0.7778 \approx 77.78\%$$